



ELITE™ AT 6501

Enhanced Polyethylene Resin

Overview

ELITE™ AT 6501 Enhanced Polyethylene Resin is an enhanced LLDPE ethylene-octene copolymer from Dow. It offers a unique combination of low seal initiation temperature, high hot tack strength, high impact resistance, good optical properties and it is easy to process.

Main Characteristics:

- Low seal initiation temperature and broad hot tack window
- High hot tack strength
- Extremely high impact resistance and good tear properties
- Good optical properties
- High throughput resin with excellent bubble stability

Complies with:

- U.S. FDA FCN 424
- EU, No 10/2011
- Consult the regulations for complete details.

Additive

- Antiblock: No
- Slip: No
- Processing Aid: No

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	0.914 g/cm ³	0.914 g/cm ³	ASTM D792
Melt Index (190°C/2.16 kg)	0.85 g/10 min	0.85 g/10 min	ASTM D1238
Films	Nominal Value (English)	Nominal Value (SI)	Test Method
Film Thickness - Tested	2.0 mil	50 µm	
Film Puncture Energy	85.0 in·lb	9.60 J	
Film Puncture Force	22.5 lbf	100 N	
Film Puncture Resistance	290 ft·lb/in ³	24.0 J/cm ³	
Secant Modulus			ASTM D882
2% Secant, MD	21500 psi	148 MPa	
2% Secant, TD	23400 psi	161 MPa	
Tensile Strength			ASTM D882
MD : Yield	1350 psi	9.30 MPa	
TD : Yield	1330 psi	9.20 MPa	
MD : Break	6380 psi	44.0 MPa	
TD : Break	6090 psi	42.0 MPa	
Tensile Elongation			ASTM D882
MD : Break	550 %	550 %	
TD : Break	600 %	600 %	
Dart Drop Impact	1500 g	1500 g	ASTM D1709A
Elmendorf Tear Strength			ASTM D1922
MD	590 g	590 g	
TD	1000 g	1000 g	
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Vicat Softening Temperature	216 °F	102 °C	ASTM D1525
Melting Temperature (DSC)	239 °F	115 °C	Dow Method
Optical	Nominal Value (English)	Nominal Value (SI)	Test Method
Gloss (45°)	64	64	ASTM D2457
Haze	8.10 %	8.10 %	ASTM D1003

Extrusion Notes

Fabrication Conditions:

- Die Diameter: 8 in
- Die Gap: 70 mil
- Melt Temperature: 429°F
- Output 10.4 lb/hr/in. of die circumference
- Screw Size: 3.5 in.
- Blow-Up Ratio: 2.5 to 1
- Screw Speed: 35 rpm
- Frost Line height: 37 in

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

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